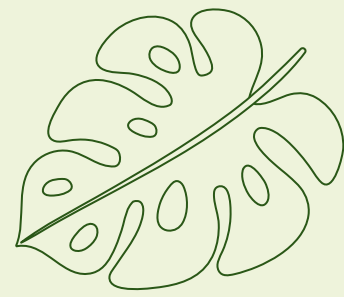




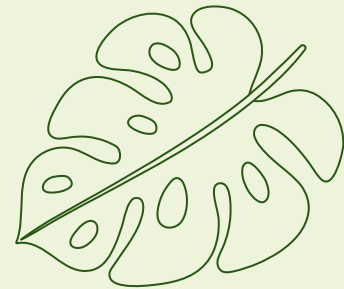
BIOPÄCK

iStart Valley 2025



Origin





The Problem

220 million tons of plastic waste annually

70 million tons ending up in the environment.

Single-use plastic packaging persists for decades in oceans and landfills

Toxins and microplastics that infiltrate food chains and water supplies.



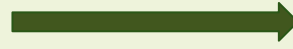
Market Statistics



**\$495.78
Billion**

Global
Biodegradable
Packaging

Projected to Reach



**\$921.25
Billion**

Due to rise in
demand for
sustainable
packaging

Food and Beverages account for



42%

Of this market



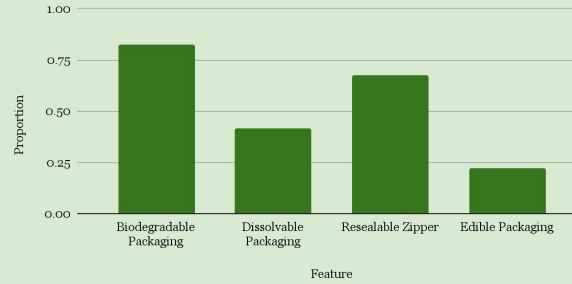
Customer Validation



How likely are you to use this product/service regularly?
(1 = very unlikely, 5 = very likely)



Which of these features would be most valuable to you?
(Respondents may select multiple)



How would you describe your current pain point with plastic pollution?
(1 = very problematic, 5 = not a problem)



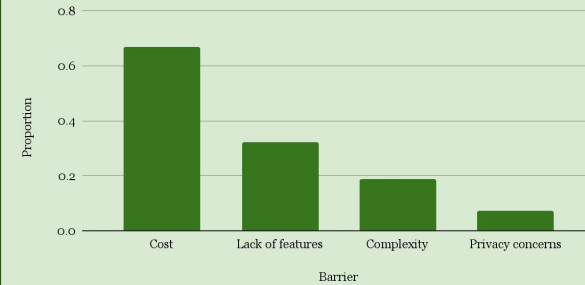
How much would you be willing to pay for a solution like this
(10 bags)?



How much does this product compare to other solutions you've
tried? (1 = much worse, 5 = much better)



What would prevent you from using this product?
(Respondents may select multiple)



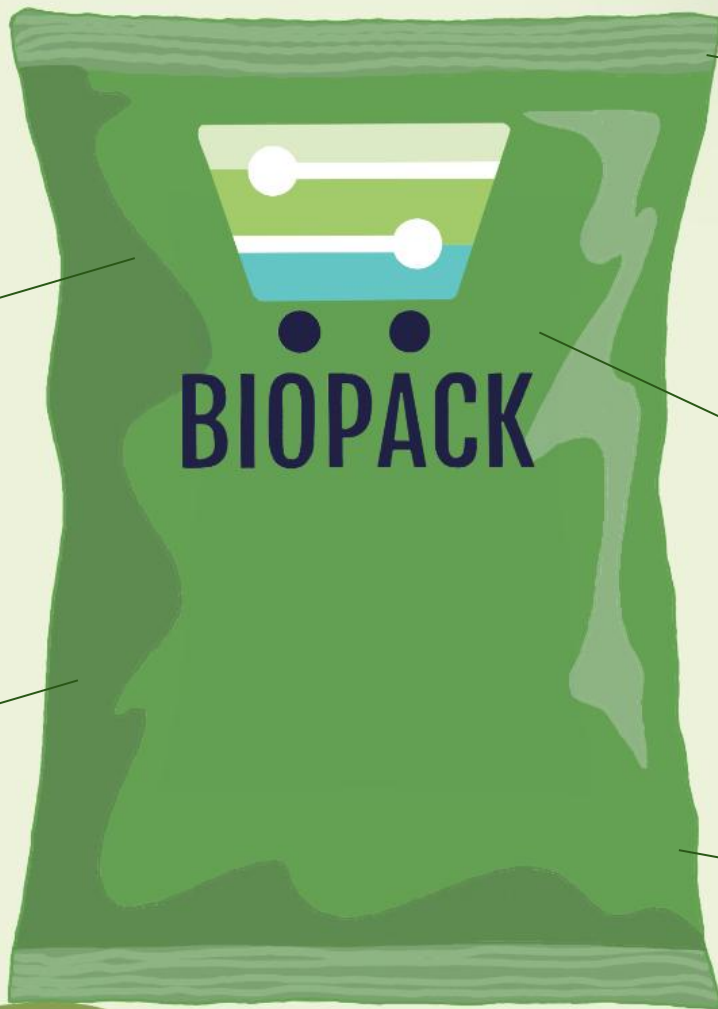
Overview of Solution

Edible

Resealable

Dissolvable

Compostable



❖ Beeswax made sealant

❖ Corn Starch base

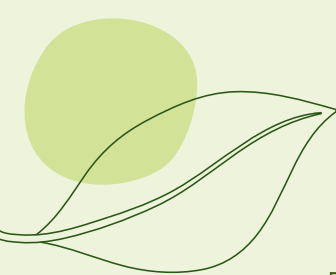
❖ Food grade citric acid or agar based polymers for water solubility agent

❖ Binded with natural gums (pectin and guar)

❖ Food grade citric acid or agar based polymers for water solubility agent

General Costs (given \$100k evaluation)

Category	Specifics	Cost Estimate
Material R&D and Prototyping	Testing corn starch, pectin, gums, beeswax formulations	\$15,000
Small-scale Manufacturing Setup	Biodegradable packaging requires food-safe production standards	\$30,000
Certifications	Compostable, food safety, biodegradable certifications	\$5,000
Marketing and Branding	Eco-friendly brand image targeting Gen Z/Millennials	\$8,000
Website and E-commerce Setup	Shopify/WordPress site, digital storefront	\$3,000
Packaging and Logistics	Food-grade handling for deliveries	\$5,000
Admin/Legal/IP	Incorporation, trademark, and basic IP protection	\$5,000
Team (expanding)	Part-time marketer, production support, consultant	\$15,000



Financial Projections



Year	Revenue	COGS	Gross Profit	Net Income
1	\$40,000	\$25,000	\$15,000	-\$5,000
2	\$150,000	\$75,000	\$75,000	\$20,000
3	\$400,000	\$180,000	\$220,000	\$100,000

Year 1

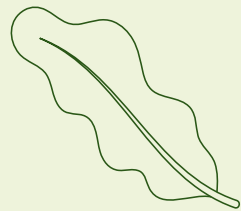
Pilot sales; focus on eco-conscious niches (decor, gifts, boutique food chains)

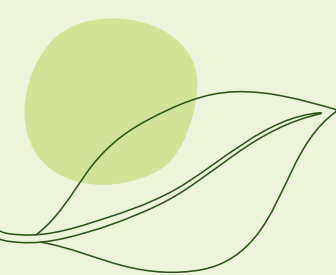
Year 2

B2B contracts with cafes, event organizers

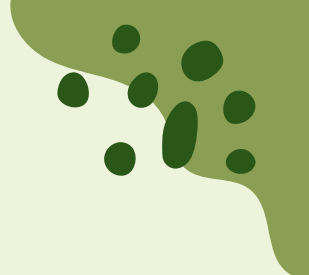
Year 3

Scale to online retail, subscription packaging options





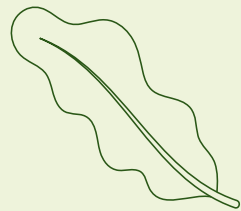
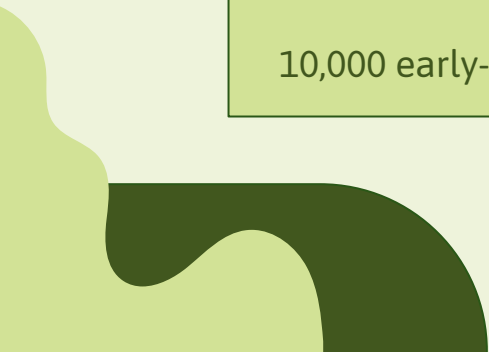
TAM/SAM/SOM Estimates



Metric	Description	Estimated Value
TAM	Global food + beverage sustainable packaging	~\$921B projected
SAM	Biodegradable/edible packaging within U.S.	~\$40-60B
SOM	Initial reachable market (decor, gifts, eco-restaurants)	~\$2-4M

SOM Detail:

10,000 early-adopting businesses × \$300/year spend = ~\$3M reachable in 2-3 years

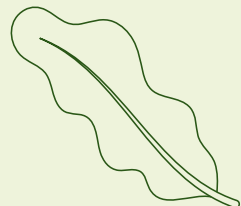




Other Costs for Improvements



Pilot product giveaways: \$2,000–\$5,000 (customer acquisition)
Event partnerships (eco expos, farmers' markets): \$1,000–\$3,000 per event
Technology upgrades: for water solubility packaging improvements
Insurance: for edible product liability



Competitors

	Amcor	Notpla	Footprint
Strengths	- Global reach - R&D power -innovative packaging	- Seaweed innovation - Eco branding - Decompose within week	- Plant-based fiber - Fast food partnerships
Weaknesses	- Still plastic-heavy - Not fully compostable	- High cost - Hard to scale	- Industrial composting required - Expensive
Opportunities	- Shift toward sustainability	- Plastic-free demand rising	- Market growth in plastic bans
Threats	- Green regulations - Rising green startups	- Material limits - Price instability	- Supply chain risks - Intense competition

Advantage

Feature	BioPack Advantage
Fully Biodegradable	Home compostable, unlike Amcor or Footprint
Cost-Efficient	Cheaper to produce than Notpla or Footprint
Scalable	Flexible design + local sourcing = global potential
Regulation-Ready	Already aligned with future green laws
Broad Market Fit	Ideal for food, retail, delivery, and more

Our Team



Aditi Devarakonda



Sejal Padhy



Aarush Mane



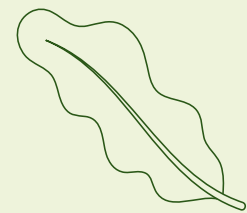
Joanna Pottipadu and Keshav krishnan

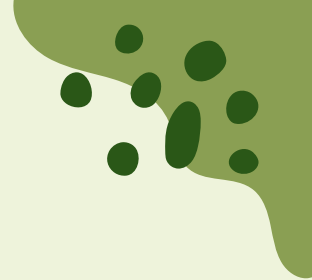




“Sustainability shouldn’t be a luxury. BioPack makes it standard.”

- Combines **eco-innovation + market scalability**
- Targets both **developing and developed markets**
- Fills the gap between **big legacy players** and **costly niche startups**





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Thank you!

Any questions?

